ExpEcon Methods: Why Incentivize??

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Why Pay?

Is there *really* a reason to pay subjects?

- I *still* get asked this pretty frequently.
- Ned Augenblick: “Why are we fetishizing incentives?”
- Danz Vesterlund Wilson (2022): Pay but don’t explain
- Enke-Graeber: Unincentivized measure of decision confidence

What do the data say?? This should be an empirical question...

The Effect of Incentives

Camerer and Hogarth [1999] remains the classic reference
They compare hypothetical, low, high payments.

1. Modal results don't change
2. ↑ payments reduce noise
3. ↑ payments induce more effort, performance
4. ↑ payments reduce desirability bias (generosity, risk-seeking)
5. Cognitive capital and costs are important, too
6. Rationality violations still persist with ↑ payments

The Effect of Incentives

Gneezy and Rustichini [2000] test various payment levels

1. IQ task
    - U-shaped performance. “Pay enough or not at all”
2. Hire HS students to soliciting money for charity
    - U-shaped performance
    - No pay > high pay > low pay

Are these tasks similar to typical experiment tasks?

The Effect of Incentives

Brañas-Garza et al. [2021]: donate x% of your lottery winnings.
High stakes $\uparrow$ total giving, but $\downarrow$ fraction, $\downarrow$ 100% giving

Ultimatum game:

- Slonim and Roth [1998]
- Andersen et al. [2011]

Hypothetical Incentives & Beliefs

Why pay for beliefs? The mechanisms are complex Danz et al. [2022]
Arguments in favor:

1. Induces subjects to take time to report truthfully
2. Might improve beliefs if belief formation is costly
   - But do we want that?? Discuss.
3. Smith's dominance [Wilde, 1981, Smith, 1982]
   - Stated beliefs used to justify selfish behavior [Blanco et al., 2010]
   - Wanting to appear more confident than they are
   - Example: salesperson

Hypothetical Incentives & Beliefs

Arguments against paying:

1. Not needed; people don't like to lie [Gneezy, 2005, Fischbacher and Föllmi-Heusi, 2013]
2. Mechanism not IC for actual people
    - Complex mechanism w/ flat maximum can crowd out intrinsic motive to report truthfully.
    - Danz et al. [2022]: calculator screws up responses

Hypothetical Incentives & Beliefs

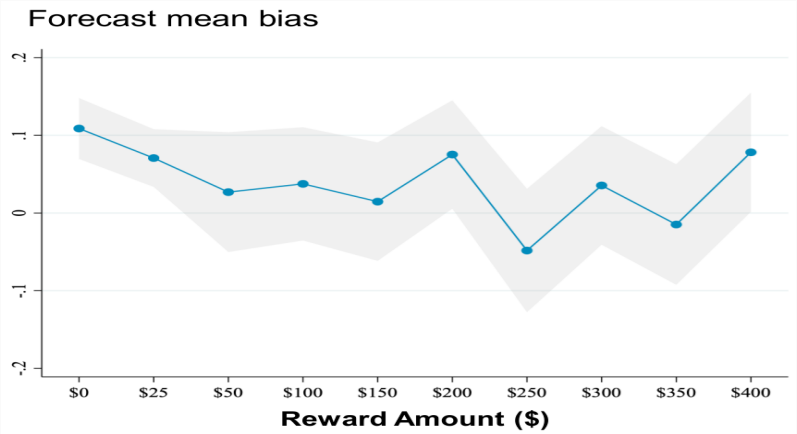
OK but what do the data say? This is a science...
Studies that show incentives improve beliefs:

- Posteriors closer to Bayes [Phillips and Edwards, 1966, Grether, 1980, Wright and Anderson, 1989]
- Burfurd and Wilkening [2022]:
  - People w/ basic grasp of Bayes's Rule: $\downarrow$ errors
  - People w/out grasp of Bayes: Update required: no difference
    No update required (uninformative signal): incentives are worse!
- Wright and Aboul-Ezz [1988]: beliefs closer to truth (eg, average GMAT scores)
- More accurate beliefs in games [Gächter and Renner, 2010, Wang, 2011]
- Harrison [2014] complex patterns of hypothetical bias
  - Paying a flat fee largely fixes it!!

Hypothetical Incentives & Beliefs

Studies that show incentives improve beliefs:

- Incentives improve belief formation
  - No incentives $\Rightarrow$ default/focal values (50% or 100% [Massoni et al., 2014, Burfurd and Wilkening, 2022]
  - And $E \succ E^C$ yet $p(E) < 1/2$ [Grether, 1992]
- Incentives reduce noise
  - Camerer and Hogarth [1999], Gächter and Renner [2010], and Trautmann and van de Kuilen [2015]. Paying for power!
- Higher incentives reduce overconfidence
  - Bloom et al. [2025]: firms guess future revenue
  - Paid \$$x$ if guess is within $\pm 10\%$ (what does that elicit?)



Hypothetical Incentives & Beliefs

Studies that show no or even negative effect of incentives:

- Sonnemans and Offerman [2001] and Trautmann and van de Kuilen [2015]
- BDM vs. Unincentivized
  - Massoni et al. [2014]: tie
  - Hollard et al. [2016]: BDM $\succ$ no pay
- Armantier and Treich [2013] incentives are worse, but could be due to risk aversion
- Trautmann and van de Kuilen [2015]: look at $p(E) + p(E^C) = 1$
  More often true *without* incentives.

Suggested Reading

Colin F. Camerer and Robin M. Hogarth. The Effects of Financial Incentives in Experiments: A Review and Capital-Labor Production Framework. *Journal of Risk and Uncertainty*, 19:7–42, 1999.

Uri Gneezy and Aldo Rustichini. Pay enough or don't pay at all. *The Quarterly journal of economics*, 115(3):791–810, 2000.

Pablo Brañas-Garza, Diego Jorrat, Jaromír Kovářík, and María C. López. Hyper-altruistic behavior vanishes with high stakes. *PLoS ONE*, 16(8):e0255668, August 2021. ISSN 1932-6203. doi: 10.1371/journal.pone.0255668.

Robert Slonim and Alvin E. Roth. Learning in High Stakes Ultimatum Games: An Experiment in the Slovak Republic. *Econometrica*, 66(3):569–596, 1998. ISSN 0012-9682. doi: 10.2307/2998575.

Steffen Andersen, Seda Ertaç, Uri Gneezy, Moshe Hoffman, and John A. List. Stakes Matter in Ultimatum Games. *American Economic Review*, 101(7):3427–3439, December 2011. ISSN 0002-8282. doi: 10.1257/aer.101.7.3427.

Suggested Reading

David Danz, Lise Vesterlund, and Alistair J. Wilson. Belief Elicitation and Behavioral Incentive Compatibility. *American Economic Review*, 112(9):2851–2883, September 2022. ISSN 0002-8282. doi: 10.1257/aer.20201248.

Louis L. Wilde. On the Use of Laboratory Experiments in Economics. In Joseph C. Pitt, editor, *Philosophy in Economics: Papers Deriving from and Related to a Workshop on Testability and Explanation in Economics Held at Virginia Polytechnic Institute and State University, 1979*, pages 137–148. Springer Netherlands, Dordrecht, 1981. ISBN 978-94-009-8394-6. doi: 10.1007/978-94-009-8394-6_9.

Vernon L. Smith. Microeconomic Systems as an Experimental Science. *The American Economic Review*, 72(5):923–955, 1982. ISSN 0002-8282.

Mariana Blanco, Dirk Engelmann, Alexander K. Koch, and Hans-Theo Normann. Belief elicitation in experiments: Is there a hedging problem? *Experimental Economics*, 13:412–438, 2010.

Uri Gneezy. Deception: The Role of Consequences. *American Economic Review*, 95(1): 384–394, February 2005. ISSN 0002-8282. doi: 10.1257/0002828053828662.

Suggested Reading

Urs Fischbacher and Franziska Föllmi-Heusi. Lies in Disguise—An Experimental Study on Cheating. *Journal of the European Economic Association*, 11(3):525–547, June 2013. ISSN 1542-4766. doi: 10.1111/jeea.12014.

Lawrence D. Phillips and Ward Edwards. Conservatism in a simple probability inference task. *Journal of Experimental Psychology*, 72(3):346–354, 1966. ISSN 0022-1015. doi: 10.1037/h0023653.

David M. Grether. Bayes Rule as a Descriptive Model: The Representativeness Heuristic. *The Quarterly Journal of Economics*, 95(3):537–557, November 1980. ISSN 0033-5533. doi: 10.2307/1885092.

William F Wright and Urton Anderson. Effects of situation familiarity and financial incentives on use of the anchoring and adjustment heuristic for probability assessment. *Organizational Behavior and Human Decision Processes*, 44(1):68–82, August 1989. ISSN 0749-5978. doi: 10.1016/0749-5978(89)90035-6.

Ingrid Burfurd and Tom Wilkening. Cognitive heterogeneity and complex belief elicitation. *Experimental Economics*, 25(2):557–592, April 2022. ISSN 1573-6938. doi: 10.1007/s10683-021-09722-x.

Suggested Reading

William F Wright and Mohamed E Aboul-Ezz. Effects of extrinsic incentives on the quality of frequency assessments. *Organizational Behavior and Human Decision Processes*, 41(2):143–152, April 1988. ISSN 0749-5978. doi: 10.1016/0749-5978(88)90023-4.

Simon Gächter and Elke Renner. The effects of (incentivized) belief elicitation in public goods experiments. *Experimental Economics*, 13(3):364–377, September 2010. ISSN 1573-6938. doi: 10.1007/s10683-010-9246-4.

Stephanie W. Wang. Incentive effects: The case of belief elicitation from individuals in groups. *Economics Letters*, 111(1):30–33, April 2011. ISSN 0165-1765. doi: 10.1016/j.econlet.2010.11.045.

Glenn W. Harrison. Hypothetical surveys or incentivized scoring rules for eliciting subjective belief distributions. *Center for Economic Analysis of Risk Working Paper Series, Robinson College of Business, Georgia State University*, 2014.

E. Elisabet Rutström and Nathaniel T. Wilcox. Stated beliefs versus inferred beliefs: A methodological inquiry and experimental test. *Games and Economic Behavior*, 67 (2):616–632, November 2009. ISSN 0899-8256. doi: 10.1016/j.geb.2009.04.001.

Suggested Reading

Sébastien Massoni, Thibault Gajdos, and Jean-Christophe Vergnaud. Confidence measurement in the light of signal detection theory. *Frontiers in Psychology*, 5, December 2014. ISSN 1664-1078. doi: 10.3389/fpsyg.2014.01455.

David M. Grether. Testing bayes rule and the representativeness heuristic: Some experimental evidence. *Journal of Economic Behavior & Organization*, 17(1):31–57, January 1992. ISSN 0167-2681. doi: 10.1016/0167-2681(92)90078-P.

Stefan T. Trautmann and Gijs van de Kuilen. Belief Elicitation: A Horse Race among Truth Serums. *The Economic Journal*, 125(589):2116–2135, December 2015. ISSN 0013-0133. doi: 10.1111/ecoj.12160.

Nicholas Bloom, Mihai A. Codreanu, and Robert A. Fletcher. Rationalizing Firm Forecasts, January 2025.

Joep Sonnemans and Theo Offerman. Is the Quadratic Scoring Rule really incentive compatible? Technical report, Citeseer, 2001.

Guillaume Hollard, Sébastien Massoni, and Jean-Christophe Vergnaud. In search of good probability assessors: An experimental comparison of elicitation rules for confidence judgments. *Theory and Decision*, 80(3):363–387, March 2016. ISSN 1573-7187. doi: 10.1007/s11238-015-9509-9.

Suggested Reading

Olivier Armantier and Nicolas Treich. Eliciting beliefs: Proper scoring rules, incentives, stakes and hedging. *European Economic Review*, 62:14–40, 2013.